SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0709

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology . (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I **R.Varshini / 1921260207** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R.Varshini

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

STUDENT DETAILS

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Course Code	:	CSA0709
Course Name	:	Computer Network
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Question 1 – Smart Campus Subnet Allocation

Given: Block 192.168.100.0/24 (256 addresses). School of Computing /25, School of Electronics /26, School of Mechanical Engineering /27.
Approach: Subnets are allotted largest first (/25, then /26, then /27) so that they sit on correct boundaries without overlapping. Block size = $2^{(32 - \text{prefix})}$: /25 = 128, /26 = 64, /27 = 32. Usable hosts = block size - 2.

Tasks 1, 2 and 3: Network, broadcast, host range and usable hosts

Department	Subnet / Mask	Network address	Broadcast address	Valid host range	Usable hosts
School of Computing	192.168.100.0/25 255.255.255.128	192.168.100.0	192.168.100.127	192.168.100.1 to 192.168.100.126	2^7 - 2 = 126
School of Electronics	192.168.100.128/26 255.255.255.192	192.168.100.128	192.168.100.191	192.168.100.129 to 192.168.100.190	2^6 - 2 = 62
School of Mechanical Engg.	192.168.100.192/27 255.255.255.224	192.168.100.192	192.168.100.223	192.168.100.193 to 192.168.100.222	2^5 - 2 = 30

Task 4: Remaining unused range

Addresses used = 128 + 64 + 32 = 224. Addresses left = 256 - 224 = 32.
Unused range: 192.168.100.224 to 192.168.100.255 (i.e. 192.168.100.224/27, mask 255.255.255.224). If used later as a subnet, network = 192.168.100.224, broadcast = 192.168.100.255, hosts = 192.168.100.225 - 192.168.100.254 (30 usable).
Decision: Allocating largest-first leaves one clean /27 block free at the end of the range for future growth, with no wasted gaps between subnets.

Question 2 – Multinational Company Subnet Allocation (10.10.0.0/16)

Given: Block 10.10.0.0/16 (65,536 addresses). Software Development /24, Quality Assurance /25, Human Resources /26, Executive Management /27.
Approach: Allocate sequentially, largest to smallest, starting at 10.10.0.0. Block sizes: /24 = 256, /25 = 128, /26 = 64, /27 = 32. Usable hosts = block size - 2.

Tasks 1, 2 and 3: Network, broadcast, host range and usable hosts

Department	Subnet / Mask	Network address	Broadcast address	Valid host range	Usable hosts
Software Development	10.10.0.0/24 255.255.255.0	10.10.0.0	10.10.0.255	10.10.0.1 to 10.10.0.254	2^8 - 2 = 254
Quality Assurance	10.10.1.0/25 255.255.255.128	10.10.1.0	10.10.1.127	10.10.1.1 to 10.10.1.126	2^7 - 2 = 126
Human Resources	10.10.1.128/26 255.255.255.192	10.10.1.128	10.10.1.191	10.10.1.129 to 10.10.1.190	2^6 - 2 = 62
Executive Management	10.10.1.192/27 255.255.255.224	10.10.1.192	10.10.1.223	10.10.1.193 to 10.10.1.222	2^5 - 2 = 30

Task 4: Remaining unused range

Addresses used = 256 + 128 + 64 + 32 = 480. Addresses left = 65,536 - 480 = **65,056**.
Unused range: 10.10.1.224 to 10.10.255.255, made up of:
• 10.10.1.224 - 10.10.1.255 (32 addresses, the rest of the 10.10.1.x block)
• 10.10.2.0 - 10.10.255.255 (65,024 addresses = 254 x 256)
Decision: All four departments fit inside the first two /24 blocks (10.10.0.x and 10.10.1.x), keeping the address plan compact and leaving 10.10.2.0 onward free for new departments or summarised routing.

Question 3 – IPv4 to IPv6 Migration (Smart Hospital)

Given: 2001:0db8:abcd:0000:0000:0000:0000:0000 /64

Task 1: Compressed notation

Rule 1: drop leading zeros in each group (0db8 becomes db8, 0000 becomes 0). Rule 2: replace the longest run of all-zero groups with **::** (once only).
2001:db8:abcd:0:0:0:0:0 becomes **2001:db8:abcd::/64**

Task 2: Expanded (full) notation

Restore the leading zeros and expand :: into the missing all-zero groups so that there are 8 groups of 4 hex digits:
2001:0db8:abcd:0000:0000:0000:0000:0000/64

Task 3: Network prefix (/64)

/64 means the first 64 bits (first 4 groups) identify the network; the last 64 bits identify the interface (host).

Part	Bits	Value
Global routing prefix (/48)	48	2001:0db8:abcd
Subnet ID	16	0000
Network prefix (/64)	**64**	**2001:0db8:abcd:0000 :: /64 (2001:db8:abcd::/64)**
Interface ID (host portion)	64	0000:0000:0000:0000 to ffff:ffff:ffff:ffff

Task 4: Number of host addresses in the subnet

Host bits = 128 - 64 = 64, so addresses = 2^{64} = **18,446,744,073,709,551,616** (about 1.8 x 10^{19}).
IPv6 has no broadcast address, so there is no "minus 2" as in IPv4; essentially the whole range is assignable to hosts (only the all-zeros subnet-router anycast address is reserved by convention).

Task 5: Total possible host addresses available in this network

Total possible addresses in 2001:db8:abcd::/64 = **2^{64} = 18,446,744,073,709,551,616**, from 2001:db8:abcd::0 to 2001:db8:abcd:0:ffff:ffff:ffff:ffff. This is more than enough for thousands of medical devices and IoT sensors; the hospital could also create 2^{16} = 65,536 such /64 subnets from its /48 prefix.

Question 4 – ISP Core Router Forwarding

Routing table: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. **Destination IP of packet:** 192.168.2.45

Task 1: Destination subnet

The router ANDs the destination address with each entry's mask (255.255.255.0):

Route entry	Mask	192.168.2.45 AND mask	Match?
192.168.1.0/24	255.255.255.0	192.168.2.0	No (192.168.2.0 is not 192.168.1.0)
192.168.2.0/24	255.255.255.0	192.168.2.0	**Yes**
192.168.3.0/24	255.255.255.0	192.168.2.0	No (192.168.2.0 is not 192.168.3.0)

Binary check: 192.168.2.45 = 11000000.10101000.00000010.00101101; the first 24 bits (11000000.10101000.00000010) equal the network 192.168.2.0.
The packet belongs to subnet 192.168.2.0/24 (hosts 192.168.2.1 - 192.168.2.254, broadcast 192.168.2.255).

Task 2: Matching routing table entry

192.168.2.0/24. It is the only entry that matches (longest-prefix match, 24 bits).

Task 3: Next-hop network / interface

The table does not list interfaces, so the standard assumption is that each customer network is attached to its own router interface. The packet is forwarded out of the **interface connected to the 192.168.2.0/24 customer network** (for example GigabitEthernet0/1 on the core router), delivered directly or to that customer's edge router, for example 192.168.2.1. Next-hop network: **192.168.2.0/24**.

Network	Assumed exit interface
192.168.1.0/24	Gi0/0
192.168.2.0/24	Gi0/1 (used for this packet)
192.168.3.0/24	Gi0/2

Task 4: Default route

If the destination matches no entry (for example 8.8.8.8), the router uses the default route **0.0.0.0/0** (mask 0.0.0.0), which matches every address with the shortest possible prefix and is used last. It points to the ISP upstream or next-hop router. Cisco configuration:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 <upstream-next-hop-IP or exit-interface>
```

Question 5 – Two LANs Connected by a Layer-3 Router

Given: Administration LAN 192.168.10.0/26 (gateway 192.168.10.1); Finance LAN 192.168.10.64/26 (gateway 192.168.10.65). A /26 mask is 255.255.255.192, block size 64, usable hosts 62.

Tasks 1 and 2: Network, broadcast and valid host range

LAN	Network address	Broadcast address	Valid host range	Usable hosts
Administration	192.168.10.0	192.168.10.63	192.168.10.1 - 192.168.10.62	62
Finance	192.168.10.64	192.168.10.127	192.168.10.65 - 192.168.10.126	62

Task 3: IP addresses for three computers in each LAN

LAN	Computer	IP address	Subnet mask	Default gateway
Administration	PC1	192.168.10.2	255.255.255.192	192.168.10.1
Administration	PC2	192.168.10.3	255.255.255.192	192.168.10.1
Administration	PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance	PC4	192.168.10.66	255.255.255.192	192.168.10.65
Finance	PC5	192.168.10.67	255.255.255.192	192.168.10.65
Finance	PC6	192.168.10.68	255.255.255.192	192.168.10.65

The router's own interface addresses (.1 and .65) are excluded; all chosen addresses lie within the valid host range of their LAN and are not the network or broadcast address.

Task 4: Default gateway for each LAN

• Hosts in the Administration LAN use **192.168.10.1** (router interface in 192.168.10.0/26).
• Hosts in the Finance LAN use **192.168.10.65** (router interface in 192.168.10.64/26).
A host sends traffic for the other LAN to its own gateway; the Layer-3 router then routes it between the two subnets. For example, 192.168.10.2 pinging 192.168.10.66 goes to 192.168.10.1, is routed, and leaves via 192.168.10.65 to reach Finance.

Router interface configuration:

```
interface Gi0/0
 ip address 192.168.10.1 255.255.255.192
 no shutdown
interface Gi0/1
 ip address 192.168.10.65 255.255.255.192
 no shutdown
```